

Objective: The goal of this project is to design and develop a simple command-line program. The Small Shop Helper program will assist shop staff in quickly calculating the total cost of items, applying discounts, and displaying the final amount clearly. Resulting in increased efficiency and reduction in calculation error as research shows that 35% of companies report discrepancies in their most important spreadsheets. All packaged in an easy-to-use system for users with minimal technical experience.

Title: Small Shop Helper Program Proposal

1. Purpose

The Small Shop Helper program will assist shop staff in efficiently calculating totals and applying discounts at checkout. The program will:

- *Add up the prices of **multiple items** quickly and accurately.*
- *Apply **discounts** based on shop promotions.*
- *Display a clear and easy-to-read **final total** for the user.*
- Reduce human error in manual calculations and save staff time during busy periods.

This simple tool will serve as a reliable and practical solution for small retail environments.

2. Target Audience

Shop staff working in small, local shops with minimal technical experience. An example persona is outlined below:

- **Age:** 30
- **Background:** Works part-time or full-time in retail; basic familiarity with computers.
- **Goals:** Quickly calculate total prices for customer purchases and apply discounts correctly.
- **Motivations:** Wants a tool that simplifies checkout and reduces calculation errors at minimal costs.
- **Pain Points:** Manual calculations are time consuming and prone to mistakes; existing tools may be too complex, expensive and/or not tailored to their needs.
- **Preferred Work Style:** Simple, step-by-step command-line instructions with clear output.

3. Design Concept

- **Layout:** Clean, text-based command-line interface. Each step prompts the user clearly for input.
- **Navigation:** Enter items > apply discount > view total. Minimal commands required.
- **Interactivity:** Python-based input prompts allow users to enter item prices, choose discount percentages, and see results instantly.
- **Accessibility:** High-contrast terminal text, simple prompts, and clear labeling to avoid confusion.

4. Scope and Assumptions

Scope:

- Enter multiple item prices.
- Apply a single discount percentage to the total.
- Display subtotal, discount amount, and final total clearly.
- Handle incorrect input (e.g., letters instead of numbers) with user-friendly error messages.

Exclusions:

- No integration with external databases or inventory systems.
- No graphical interface; strictly command-line based.
- Advanced discount schemes (e.g., “buy one get one free”) are not included.

5. Tools and Technologies

- **Python** for program development.
- Basic terminal for running the program.
- Text editor **Visual Studio Code** for coding.

6. Timeline

- **Discovery:** 1-2 days to research similar programs and confirm requirements.
- **Design:** 2-3 days to plan program flow, variables, and error handling.
- **Build:** 3-4 days to code the program, including testing for errors.
- **Test:** 1-2 days to run sample calculations and ensure accuracy.
- **Presentation:** 1 day to prepare demonstration and documentation.

KEY NOTE: THIS TIMELINE IS A STANDARD APPROXIMATION AND MAY VARY DEPENDING ON BUGGING AND TESTING NEEDS.

7. Risks and Mitigations

Risk: Users may enter invalid data (letters instead of numbers).

Mitigation: Include input validation such as `int()` and `.isnumeric` and clear error messages to prompt correct input.

Risk: Staff may find the command-line interface confusing.

Mitigation: A simple user guide and clear step-by-step instructions will be included in the program.

8. Research & Idea Sources

The ideation of this proposal came from researching some potential competitors and the issues small businesses may face. I listed these sources below:

Shopify: Ecommerce API that offers quick calculations at a monthly fee

Pros: Graphical interface with a plethora of features

Cons: Monthly Subscription

Link: <https://www.shopify.com/uk>

Sage: An accounting and financial management tool

Pros: Graphical interface with human assistance

Cons: Higher Monthly Subscription

Link: <https://www.sage.com/en-gb/blog/speed-up-business-processes/>

9. Conclusion

This proposal outlines a simple, practical, and efficient Small Shop Helper program. Once developed, it will streamline the checkout process, reduce calculation errors, and save time for shop staff, while remaining easy to use and accessible to users with little technical experience.